What means GMO?

Genetically Modified Organisms

- Recombinant DNA technology goes beyond traditional cross breeding techniques, **making possible an exchange of traits from different species, even among plants, animals and bacteria.**

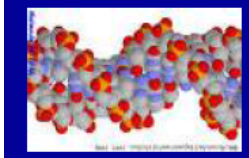
- GMO are a result of technology that has altered the DNA of living organisms (animals, plants or bacteria)

- Other terms that mean the same thing:
  - Genetically engineered
  - Transgenic
  - Recombinant DNA (rDNA) technology

Biotechnology, defined broadly, is the engineering of organisms for useful purposes.

Often, biotechnology involves the creation of hybrid genes and their introduction into organisms in which some or all of the gene is not normally present.

Recombinant DNA Technology



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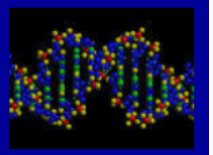


How to Mess with DNA for Fun and Profit

What the heck is Recombinant DNA?

Recombinant DNA is what you get when you combine DNA from two different sources.

For example:

Mouse + Human DNA

Human + Bacterial DNA

Viral + Bacterial DNA

Human + (other) Human DNA

Why Make Recombinant DNA?

Recombinant DNA Technology May Allow Us To:

- *Cure or treat disease*
- *Genetically modify our foods to increase flavor, yield, nutritional value or shelf-life*
- *Better understand human genetics*
- *Clone cells or organs*

Molecular Biology's Best Friends: *Bacteria*

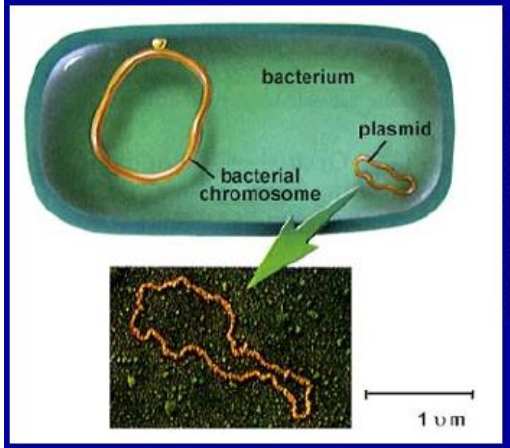
Why use bacteria?

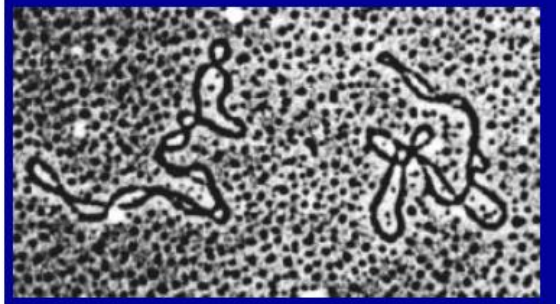
- They're relatively ***simple*** organisms.
- They reproduce ***very quickly*** and ***asexually*** (this means that the "daughter" cells will contain the exact same DNA as the "parent" cell).
- It's pretty easy to get DNA back into the bacteria after you've changed it.
- We can mess around with their DNA and kill a lot of them during our experiments and nobody gets mad. ;-)

Few vocabulary →

Plasmids

Small, circular pieces of "extra" DNA found in bacteria.





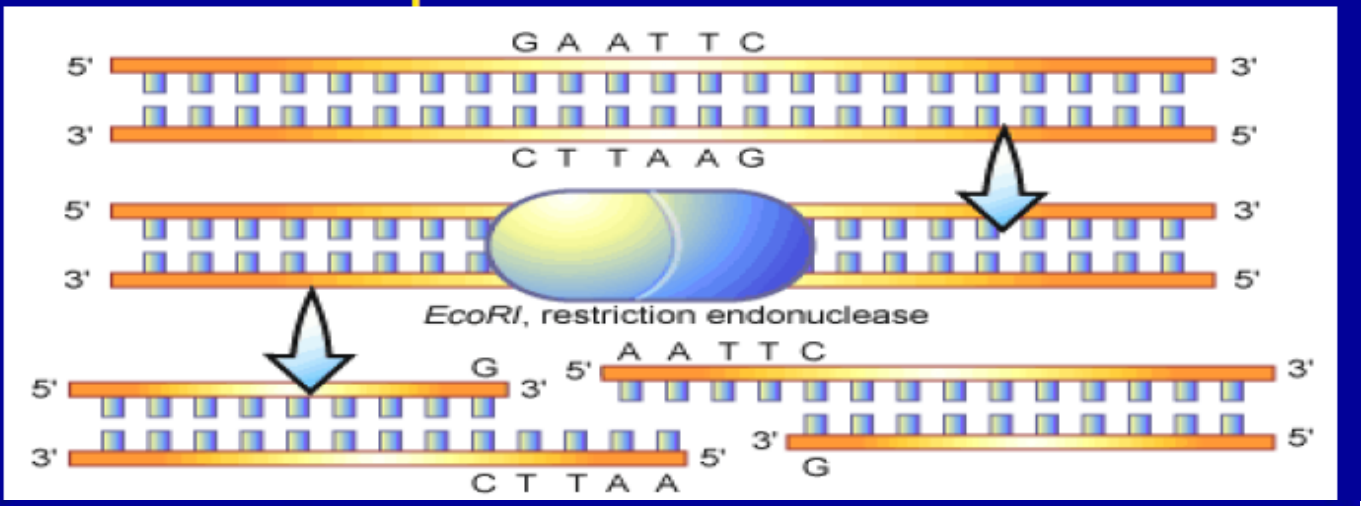
Plasmids often carry antibiotic resistance.

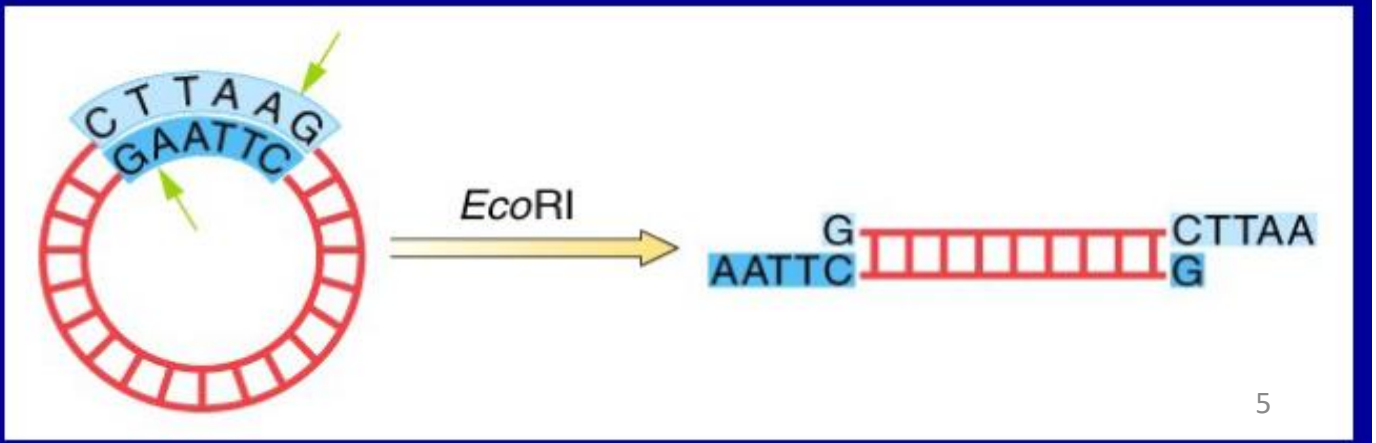
Restriction Enzymes: Molecular Scissors

- Any piece of DNA cut with a certain restriction enzyme will "stick" to any other piece cut with that same RE, even if they come from different sources.

A restriction enzyme (RE) is a specialized protein that cuts DNA in a *very specific* place.

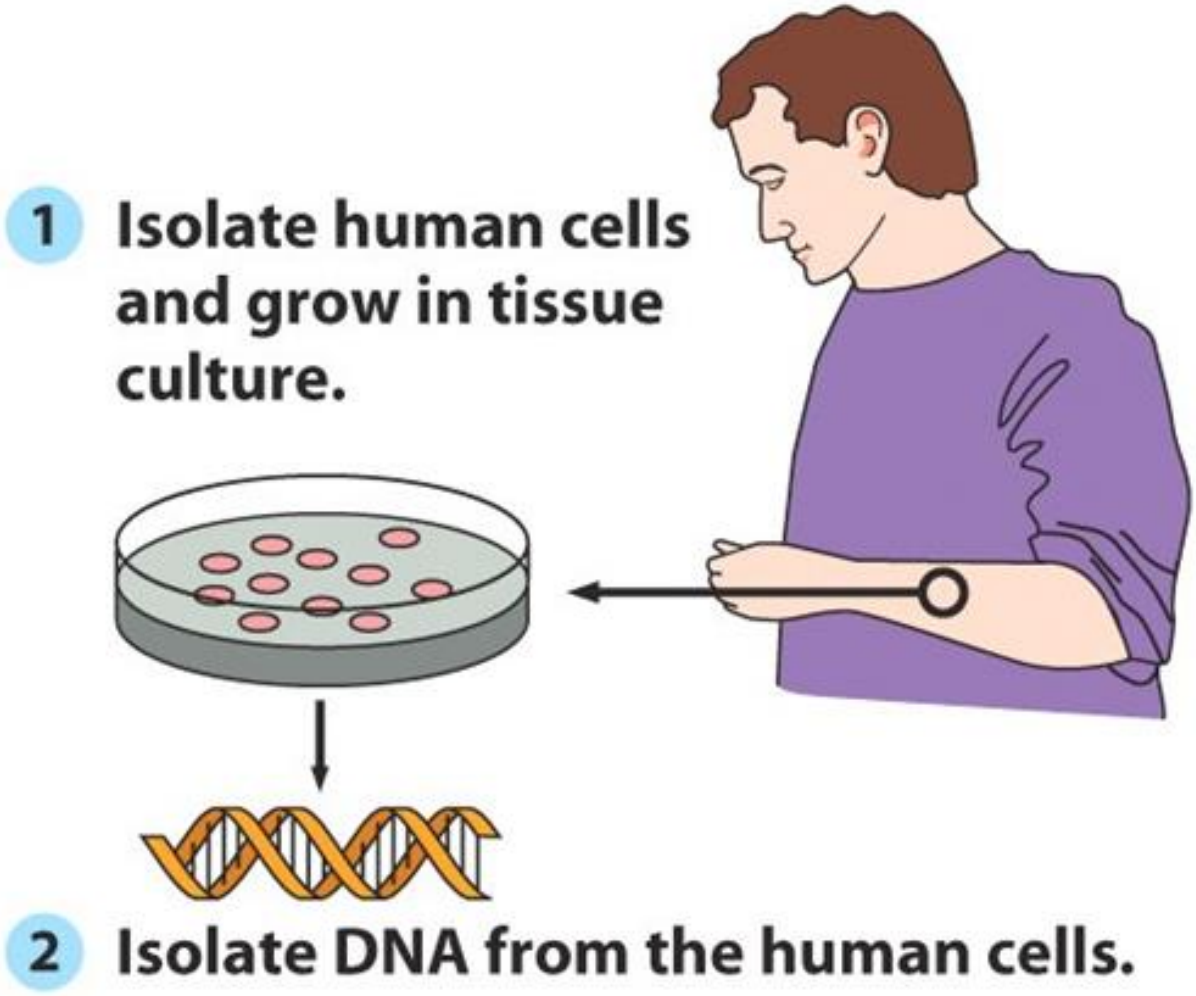
- Different REs cut at different places along the nucleotide sequence.

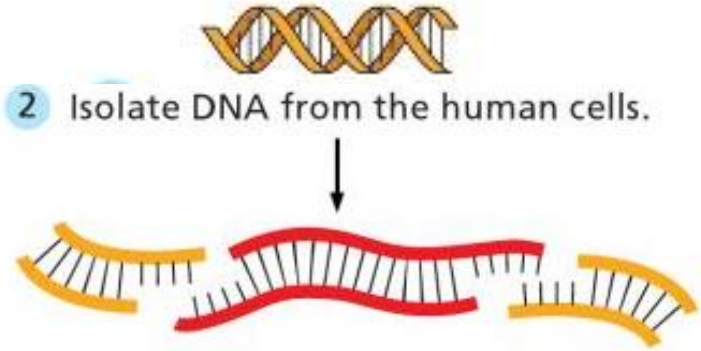




Bacterial species/strain	Enzyme name	Recognition sequences and cleavage sites
***B**acillus **amy**loliquefaciens* **H**	*Bam* H1	GGATCC CCTAGG
E**scherichia **coli Ry13	*Eco* R1	GAATTC CTTAAG
***P**rovidencia **stu**artii* 164	*Pst* 1	CTGCAG GACGTC
***S**erratia **mar**cescens* SB	*Sma* H1	CCCGGG GGGCCC
***R**hodopseudomonas **sph**aeroides*	*Rsa* 1	GTAC CATG

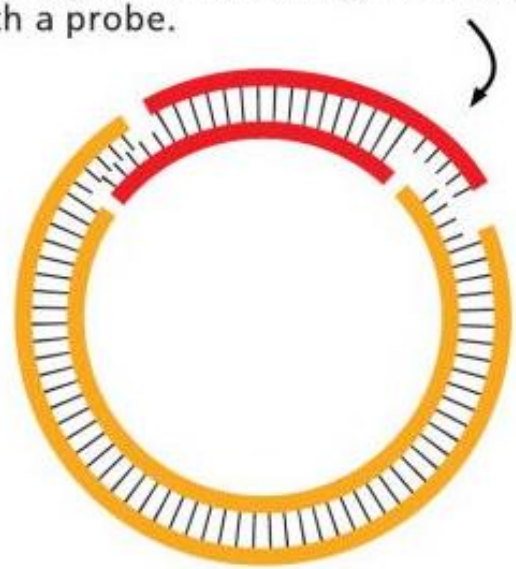
CASE STUDY: Human Insulin (Humulin) Production by Bacteria

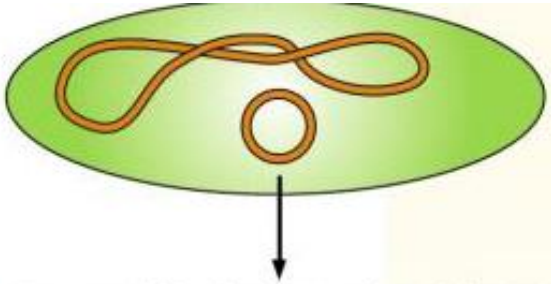




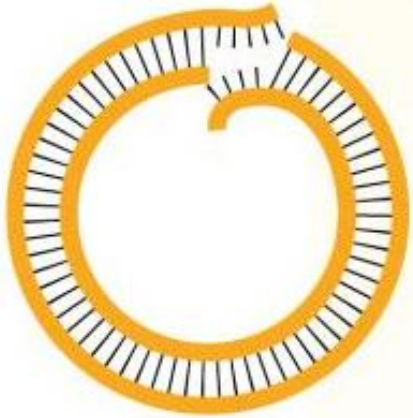
2 Isolate DNA from the human cells.

3 Use a restriction enzyme to cut DNA into fragments with sticky ends. Isolate the fragment containing 'insulin gene' with a probe.



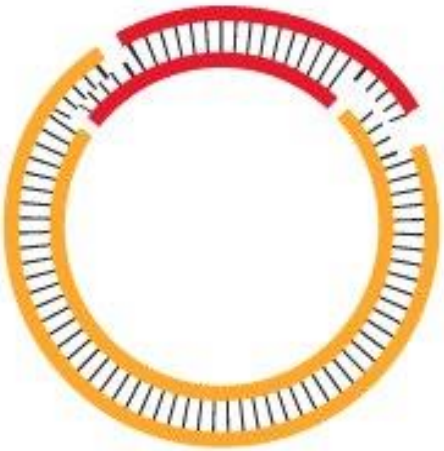


4 Meanwhile, isolate plasmid DNA from a bacterium.



5 Use the same restriction enzyme to cut the plasmid DNA, creating matching sticky ends.

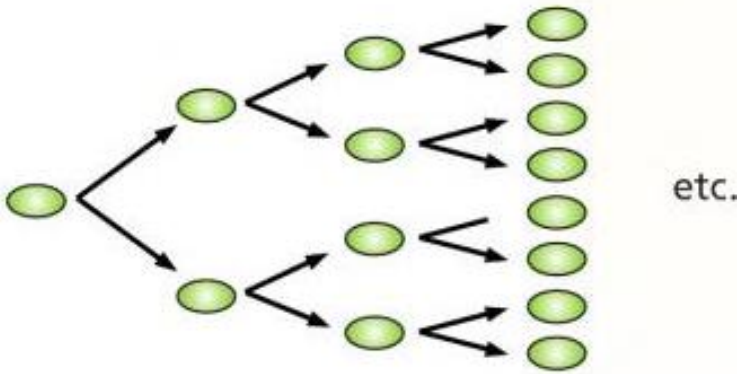
6) join the plasmid and human fragment



Mix the recombinant plasmid with bacteria.

7 Allow new bacteria to incorporate the recombinant plasmid into the bacterial cell, then screen bacteria to find the ones that have incorporated the human gene for insulin.

One cell with the recombinant plasmid



8 Grow trillions of new insulin-producing bacteria.

The final steps are to collect the bacteria, break open the cells, and purify the insulin protein expressed from the recombinant human insulin gene.

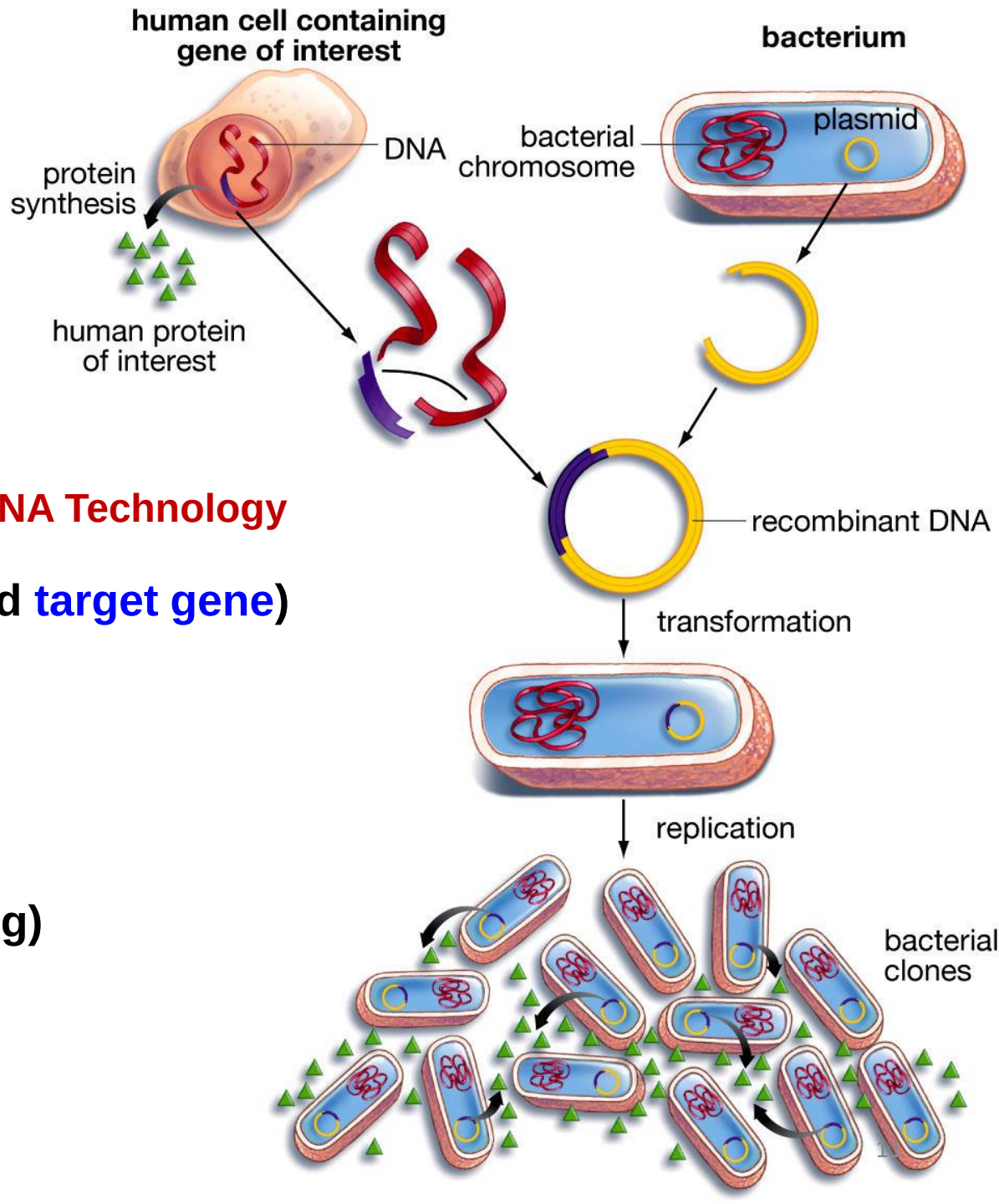
Now your E. coli will use its new DNA to make human insulin!

Because they reproduce so quickly, you'll soon have thousands, millions, or ***billions*** of human insulin making machines.

By filtering out the bacteria after they've made insulin, you've got clean human insulin that can be packaged and given to diabetic patients.

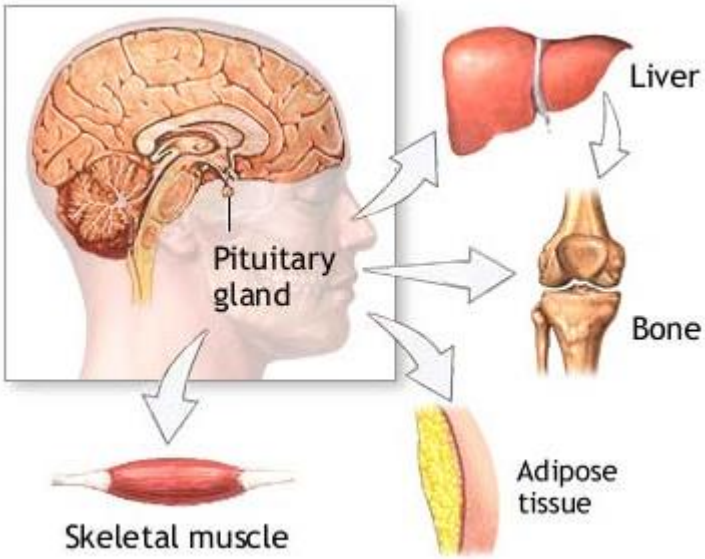
YAY!

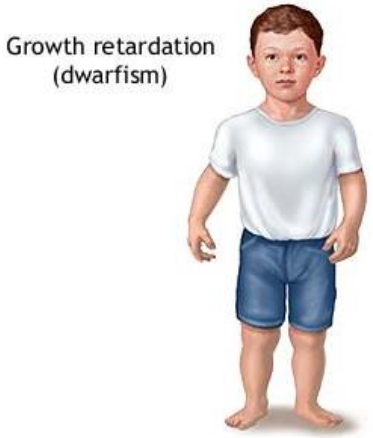
Overview: Route to the Production by Bacteria of Human Insulin



Six steps of Recombinant DNA Technology

1. Isolating (vector and target gene)
2. Cutting (Cleavage)
3. Joining (Ligation)
4. Transforming
5. Cloning
6. Selecting (Screening)





- **Insulin**
  - **Hormone required to properly process sugars and fats**
  - **Treat diabetes**
  - **Now easily produced by bacteria**

- **Growth hormone deficiency**
  - **Faulty pituitary and regulation**
  - **Had to rely on cadaver source**
  - **Now easily produced by bacteria**

SOME PROTEIN PRODUCTS OF RECOMBINANT DNA TECHNOLOGY

Product	Made In	Use
Human insulin	*E. coli*	Treatment for diabetes
Human growth hormone (GH)	*E. coli*	Treatment for growth defects
Epidermal growth factor (EGF)	*E. coli*	Treatment for burns, ulcers
Interleukin-2 (IL-2)	*E. coli*	Possible treatment for cancer
Bovine growth hormone (BGH)	*E. coli*	Improving weight gain in cattle
Cellulase	*E. coli*	Breaking down cellulose for animal feeds
Taxol	*E. coli*	Treatment for ovarian cancer
Interferons (alpha and gamma)	*S. cerevisiae; E. coli*	Possible treatment for cancer and viral infections
Hepatitis B vaccine	*S. cerevisiae*	Prevention of viral hepatitis
Erythropoietin (EPO)	Mammalian cells	Treatment for anemia
Factor VIII	Mammalian cells	Treatment for hemophilia
Tissue plasminogen activator (TPA)	Mammalian cells	Treatment for heart attacks

Use recombinant cells to mass produce proteins

- **Bacteria**
- **Yeast**
- **Mammalian**

GMO: What is it?

- US Crops — Main food for US
  - Corn* — Cornstarch, corn syrup, corn oil
  - Soy* — Soymilk, tofu, soy lecithin
  - Canola — Canola oil
  - Sugar beets — Sugar
  - Cotton — Cottonseed oil, tampons
  - Squash — Zucchini, crooked neck
  - HI Papaya — Papaya from Hawaii

* Main ingredient of US raised cows, pigs & chickens

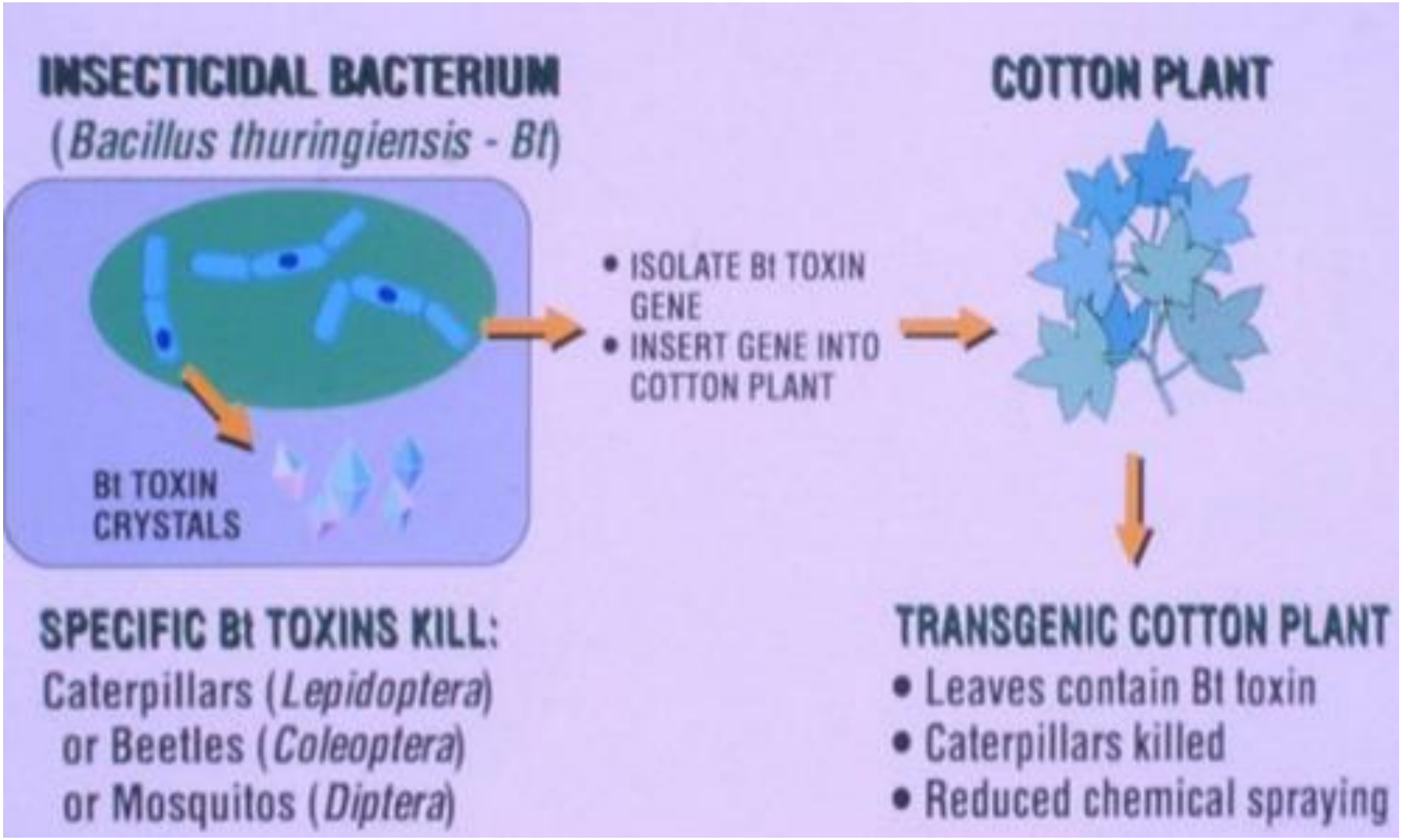
80% of what you buy in the grocery store contains GMOs

Table 19.4

Genetically Modified Crops

Altered Plant	Effect
Rice with beta carotene and extra iron	Added nutritional value
Canola with high-laurate oil	Can be grown domestically; less costly than importing palm and coconut oils
Delayed ripening tomato	Extended shelf life
Herbicide-resistant cotton	Herbicide kills weeds without harming crop
Minipeppers	Improved flavor, fewer seeds
Bananas resistant to fungal infection	Extended shelf life
Delayed-ripening bananas and pineapples	Extended shelf life
Elongated sweet pepper	Improved flavor, easier to slice
Altered cotton fiber	"Plasticized" fabric
Altered paper pulp trees	Paper component (lignin) easier to process
High-starch potatoes	Absorb less oil when fried
Pest-resistant corn	Can resist European corn borer
Seedless minimelons	Single serving size
Sweet peas and peppers	Retain sweetness longer
Sugarcane with corn gene	Resists bacterial and fungal toxins

Bt Toxin: Genetically Engineered Insecticidal Cotton



Microbial Bio-pesticide :Genetically Modified Cotton

Genetically Modified Cotton (contains a bacterial gene for pest resistance)

Standard Cotton

Advantages and Disadvantages of GMOs

Advantages	Disadvantages
Higher yield of crops due to less crops damaged by insects and disease	GMO foods don't taste as natural as non GMO foods
Less pesticides need to be purchased and used, causing more economically productive farmers and less pollution	Some organisms (such as butterflies and bees) are harmed by pesticides in GMO foods
Food production increased, causing decreasing starvation in LDCs	Labeling GMO foods causes additional expenses in processing and labeling
Added nutritional value in food due to incorporating additional vitamins into crops	Possible cross pollination of GMO foods with non GMO foods result in the inability to track GMOs
Smell of GMO foods are often enhanced	Insects can possibly evolve to be resistant to insect resistant GMO crops
GMO foods do not need as fertile land as natural foods in order to grow	Cross pollination of GMO crops with weeds could result in "super weeds"
Less additives are needed to keep GMO foods fresh	New allergies could develop to GMO foods
GMO crops are less likely to get sicknesses	Possible health complications could arise due to unknown long term affects of GMO foods
GMO crops that freeze less in the winter can be produced	GMO crops sometimes fail like normal crops (genetically modified cotton has failed in India)

Molecular Archaeology: DNA Analysis

Photograph of Iceman:
Discovered in 1991, and
approximately 5000 years old

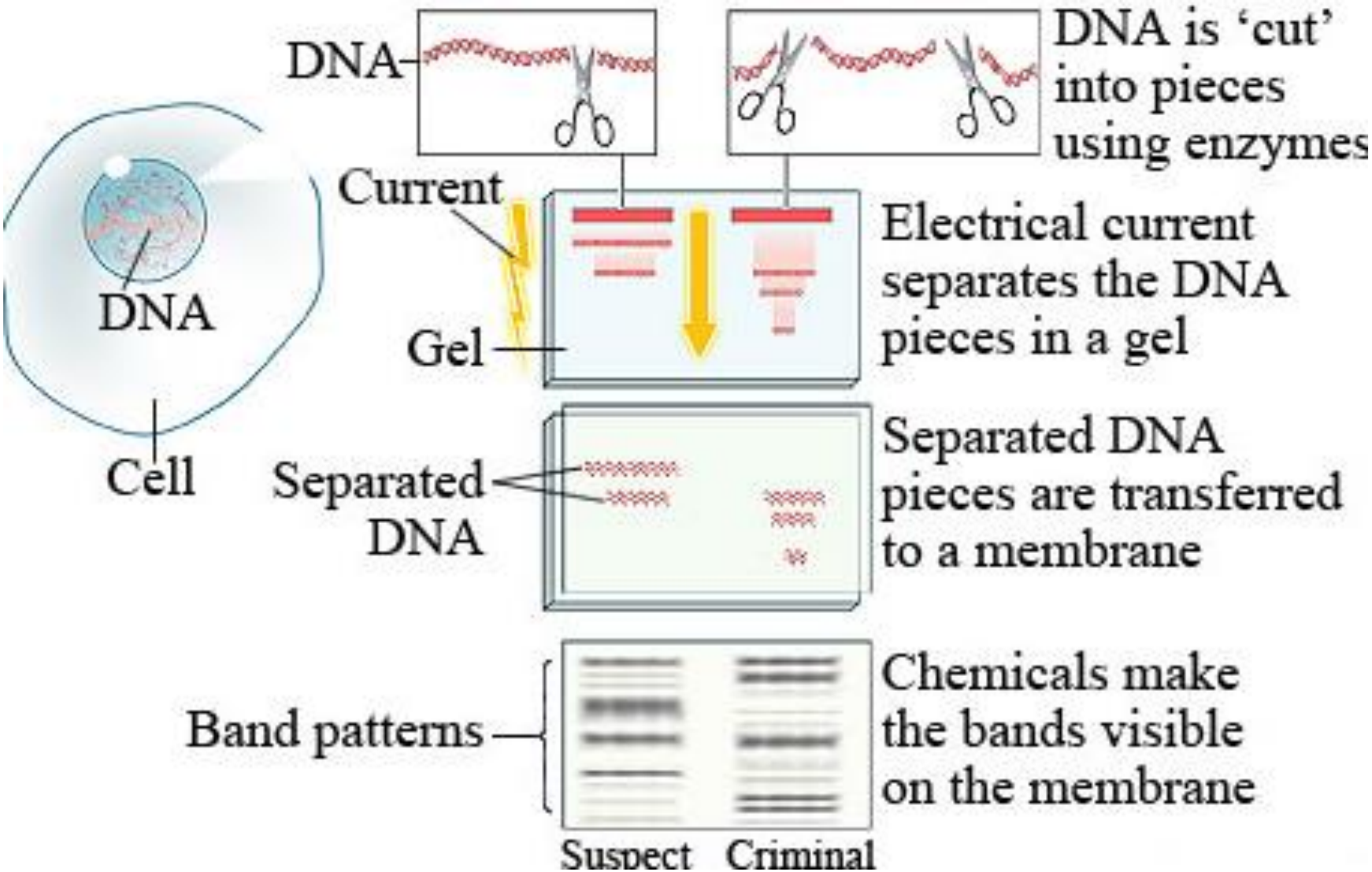
It is the oldest human sample to undergo such an analysis.
Analysis of his mitochondrial DNA indicated he came from the Italian Alps region.

DNA Analysis

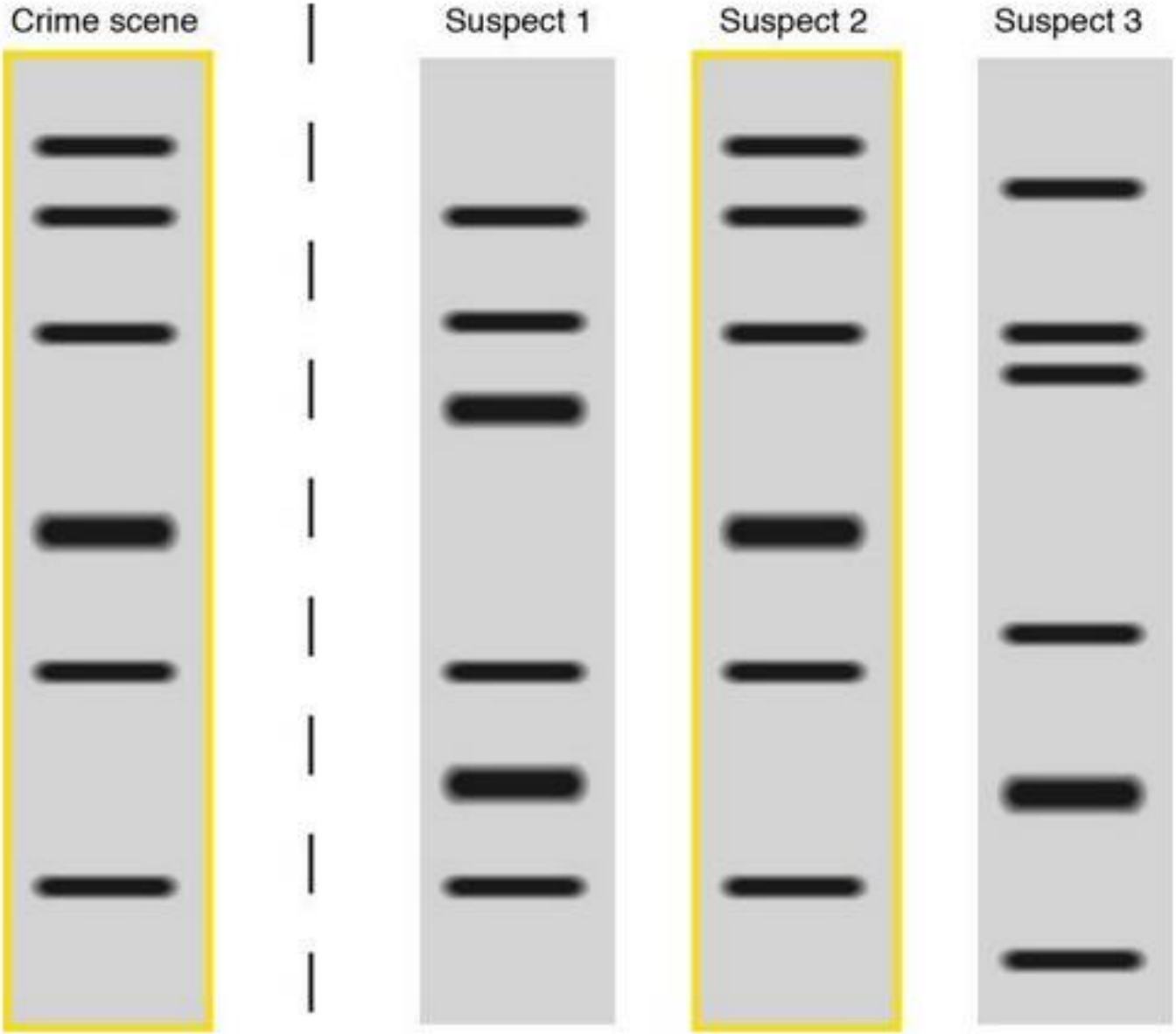
DNA Fingerprinting or DNA Typing

- Method to identify the DNA of an individual
- DNA is unique to a person, no two individuals have identical DNA except identical twins or clones
- All the cells of a person has same DNA
- A specimen for DNA fingerprinting can be any type of cell
  - Blood, saliva, sperm, muscle, teeth, bones or any DNA bearing cell

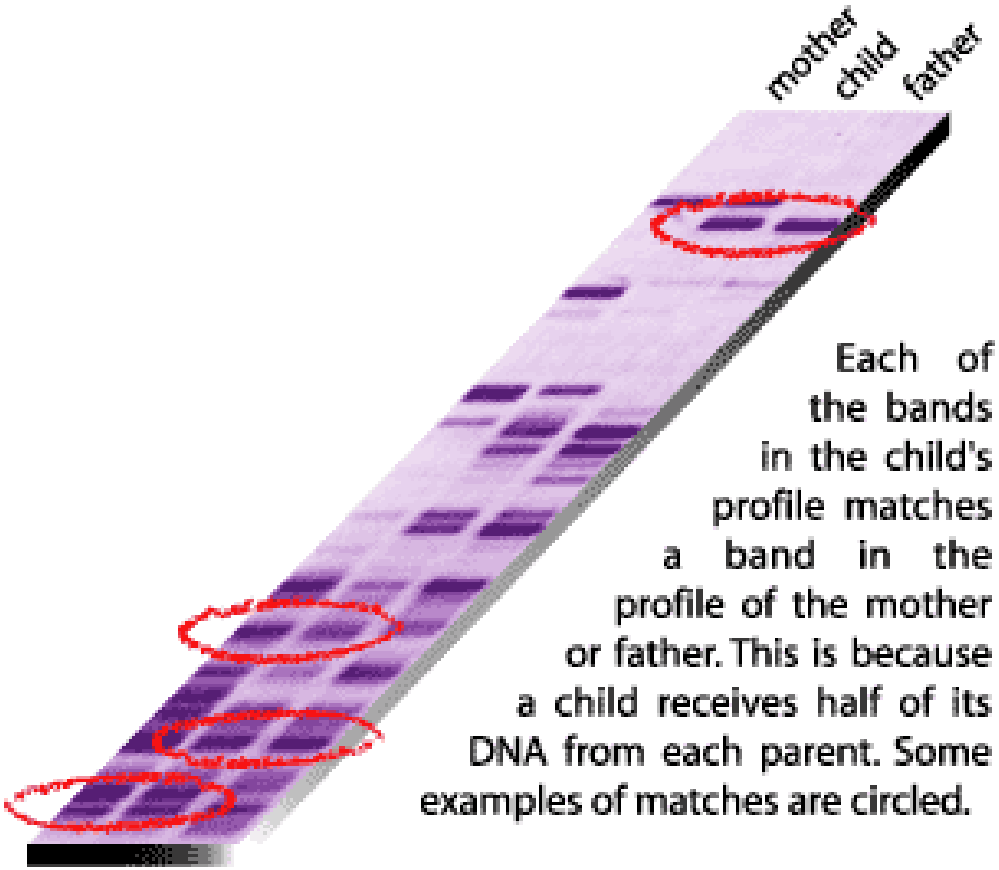
Process of Making a DNA fingerprint



How DNA fingerprinting can point to a criminal



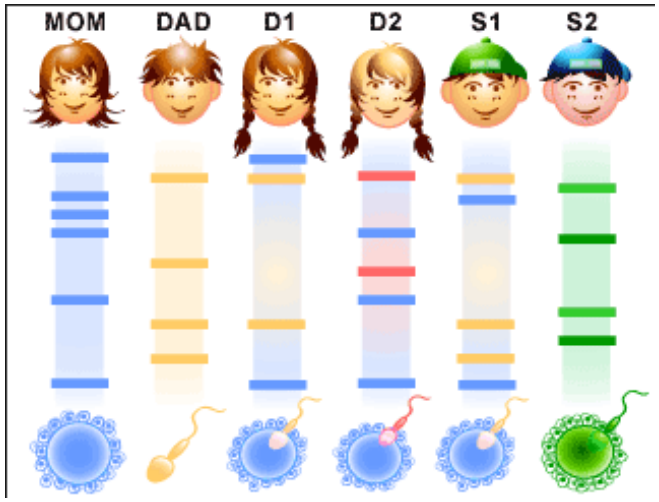
How DNA fingerprinting can identify a child's parents



Case Study: A family consists of a mom and dad, two daughters and two sons.

OBSERVATION:

Both daughter 1 and son 1 share RFLPs with both the mom and dad while daughter 2 has RFLPs of the mom but not the dad, and son 2 does not have RFLPs from either parent.



CONCLUSION:

Daughter 2 :child from the mother's previous marriage

Son 2: Adopted.